MNF130 - Oppgaver uke 6: Questions with correct answers

Based on the uploaded Task 1 PDF. Answers are added below each subquestion for practice and checking.

1. Let $P(x)$ be the statement “The word x contains the letter a”. What are these truth values?

(a) P(orange).

Answer: True. The word “orange” contains the letter a.

(b) P(lemon).

Answer: False. The word “lemon” does not contain the letter a.

(c) P(true).

Answer: False. The word “true” does not contain the letter a.

(d) P(false).

Answer: True. The word “false” contains the letter a.

2. Translate these statements into English, where $C(x)$ is “x is a comedian” and $F(x)$ is “x is funny” and the domain consists of all people.

(a) $\forall x(C(x) \rightarrow F(x))$.

Answer: Every comedian is funny.

(b) $\forall x(C(x) \land F(x))$.

Answer: Every person is a comedian and is funny.

(c) $\exists x(C(x) \rightarrow F(x))$.

Answer: There is at least one person such that if that person is a comedian, then that person is funny.

(d) $\exists x(C(x) \land F(x))$.

Answer: There is at least one person who is both a comedian and funny.

3. Let $P(x)$ be the statement “x can speak Russian” and let $Q(x)$ be the statement “x knows the computer language C++”. Express each sentence in terms of $P(x)$, $Q(x)$, quantifiers, and logical connectives. The domain consists of all students at your school.

(a) There is a student at your school who can speak Russian and who knows C++.

Answer: $\exists x(P(x) \land Q(x))$.

(b) There is a student at your school who can speak Russian but who doesn’t know C++.

Answer: $\exists x(P(x) \land \neg Q(x))$.

(c) Every student at your school either can speak Russian or knows C++.

Answer: $\forall x(P(x) \lor Q(x))$.

(d) No student at your school can speak Russian or knows C++.

Answer: $\forall x(\neg P(x) \land \neg Q(x))$. Equivalently: $\neg \exists x(P(x) \lor Q(x))$.

4. Determine the truth value of each of these statements if the domain for all variables consists of all integers.

(a) $\forall n(n^2 \geq 0)$.

Answer: True. The square of every integer is nonnegative.

(b) $\exists n(n^2 = 2)$.

Answer: False. No integer has square equal to 2.

(c) $\forall n(n^2 \geq n)$.

Answer: True. For every integer n, $n^2 - n = n(n - 1)$ is nonnegative.

(d) $\exists n(n^2 < 0)$.

Answer: False. An integer square cannot be negative.

5. Express the negation of each of these statements in terms of quantifiers without using the negation symbol.

(a) $\forall x(x > 1)$.

Answer: $\exists x(x \leq 1)$.

(b) $\forall x(x \leq 2)$.

Answer: $\exists x(x > 2)$.

(c) $\exists x(x \geq 4)$.

Answer: $\forall x(x < 4)$.

(d) $\exists x(x < 0)$.

Answer: $\forall x(x \geq 0)$.

(e) $\forall x((x < -1) \lor (x > 2))$.

Answer: $\exists x((x \geq -1) \land (x \leq 2))$.

(f) $\exists x((x < 4) \lor (x > 7))$.

Answer: $\forall x((x \geq 4) \land (x \leq 7))$.

6. Translate these specifications into English, where F(p) is "Printer p is out of service", B(p) is "Printer p is busy", L(j) is "Print job j is lost" and Q(j) is "Print job j is queued".

(a) $\exists p(F(p) \land B(p)) \rightarrow \exists jL(j)$.

Answer: If there is a printer that is out of service and busy, then there is a print job that is lost.

(b) $\forall pB(p) \rightarrow \exists jQ(j)$.

Answer: If every printer is busy, then there is a print job that is queued.

(c) $\exists j(Q(j) \land L(j)) \rightarrow \exists pF(p)$.

Answer: If there is a print job that is both queued and lost, then there is a printer that is out of service.

(d) $(\forall pB(p) \land \forall jQ(j)) \rightarrow \exists jL(j)$.

Answer: If every printer is busy and every print job is queued, then there is a print job that is lost.

7. As mentioned in the text, the notation $\exists!xP(x)$ denotes “There exists a unique x such that P(x) is true”. If the domain consists of all integers, what are the truth values of these statements?

(a) $\exists!x(x > 1)$.

Answer: False. There are many integers greater than 1.

(b) $\exists!x(x^2 = 1)$.

Answer: False. Both $x = 1$ and $x = -1$ satisfy $x^2 = 1$.

(c) $\exists!x(x + 3 = 2x)$.

Answer: True. Solving $x + 3 = 2x$ gives the unique integer $x = 3$.

(d) $\exists!x(x = x + 1)$.

Answer: False. No integer satisfies $x = x + 1$.

8. Let P(x), Q(x) and R(x) be the statements “x is a professor”, “x is ignorant” and “x is vain”, respectively. Express each of these statements using quantifiers, logical connectives, and P(x), Q(x) and R(x), where the domain consists of all people.

(a) No professors are ignorant.

Answer: $\forall x(P(x) \to \neg Q(x))$. Equivalently: $\neg\exists x(P(x) \wedge Q(x))$.

(b) All ignorant people are vain.

Answer: $\forall x(Q(x) \to R(x))$.

(c) No professors are vain.

Answer: $\forall x(P(x) \to \neg R(x))$.

(follow-up) Does (c) follow from (a) and (b)?

Answer: No. From (a), professors are not ignorant. From (b), ignorant people are vain. This does not imply that professors are not vain; a professor could be vain without being ignorant.

Note: The answer key uses x^2/n^2 notation where the PDF text extraction showed the exponent split onto a new line.